

The **OSI Model** (Open Systems Interconnection) is a **framework** that explains **how data travels** from one computer to another **in a network**.

It breaks the communication process into **7 clear layers**, each with its **own job**.

OSI Model – 7 Layers (From Top to Bottom)

Layer	Name	Role (Simple Explanation)
7	Application	What you see – software like browser, WhatsApp
6	Presentation	Formats data (like encryption, compression)
5	Session	Starts & maintains connections (sessions)
4	Transport	Ensures reliable delivery (TCP/UDP)
3	Network	Finds the path – uses IP addresses
2	Data Link	Transfers data between devices on the same network (MAC address)
1	Physical	Actual cables, signals, hardware sending bits

Easy Example (Like Sending a Letter):

Imagine you're sending a letter to a friend in another city.

Layer	Action (Letter Example)
7 App	You write the letter (browser/email)
6 Pres	You choose language, maybe encrypt it
5 Sess	You call your friend to inform you're sending it
4 Trans	You use a courier that ensures delivery (reliable or fast)
3 Network	Courier finds the best route using the address
2 Data Link	Local delivery person takes it from your house to courier office
1 Physical	The road, van, and driver actually transporting it

What Happens During Data Transfer?

When you send data:

- It goes **from Layer 7 to Layer 1** (top to bottom)
- Then reaches the other computer and goes **from Layer 1 to Layer 7** (bottom to top)

Layers with Protocols

Layer	Example Protocols
7 App	HTTP, FTP, SMTP
6 Pres	SSL, TLS
5 Sess	NetBIOS, PPTP
4 Trans	TCP, UDP
3 Network	IP, ICMP
2 D-Link	Ethernet, MAC
1 Physical	Cables, Wi-Fi, Hub



Tip to Remember Layers:

All People Seem To Need Data Processing

(Application, Presentation, Session, Transport, Network, Data Link, Physical)